

**Food Distribution Final Project**  
**IEMS 365**

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## Abstract

*This study aimed to optimize the food distribution strategy for the Food Bank of the Southern Tier through an integrated approach of demand simulation, mathematical modeling, and fairness analysis—integrating both efficiency and equity metrics to optimize service delivery to disadvantaged populations. The study undertook the systematic task of large-scale data cleaning, feature extraction using principal component analysis, and the design of mathematical programming models that account for decreasing returns and travel constraints. Techniques ranged from equitable distribution modeling that ensures fair service delivery to weather-adjusted schedule modification and resource allocation using the Eisenberg-Gale formulation, with particular emphasis on the multi-stage scheduling algorithm. Not only did the algorithm outperform the 2019 baseline in terms of overall utility by 74,279.8, but it also highlighted the challenge of ensuring equitable distribution across heterogeneous service zones as there was a gross loss of 13,712.2 in the equality gap. The findings offer actionable insights to policymakers on the fact that while strategic scheduling is able to deliver considerable efficiency gains, further application of stochastic modeling of demand and expert-based feature engineering is necessary to achieve the right balance between efficiency and fairness in food distribution.*

## 1 Introduction

Food insecurity is one of the most important social and economic issues because lack of consistent access to sufficient food for a sustainable, active life can lead to detrimental physical and mental health. Unfortunately, although this had been decreasing over the past 10 years, 2022 experienced a 45% increase in the amount of individuals that faced some sort of hunger—45 million more people than the previous year. This was primarily due to the rolling back of social security measures implemented during the Covid-19 pandemic, as well as general rising food prices.

The Food Bank of the Southern Tier (FBST) is an organization under Feeding America that attempts to provide food security to those that are economically disadvantaged in 6 New York counties. They have the Mobile Food Pantry (MFP) that provides food directly to clients in pre-packaged boxes or in a similar environment to a farmer’s market. The Covid-19 pandemic caused them to offer monthly visits to only 9 major sites, require clients to register for pick-up in advance, and use social distancing when picking up. However, they plan to no longer require clients to register in advance next year, so this study attempts to determine the optimal schedule and allocation for their 3 trucks.

In this study, several key components are

examined:

- How demand varies by location and if this correlates to geographic density or socio-economic factors
- The ideal day-to-day schedule visiting all 70 sites, while considering frequency of visits
- Optimal performance metrics for efficiency and fairness
- Accounting for the possibility of less clients on colder days
- Comparing the solution to other possible solutions

## 2 Methodology

### 2.1 Data Sources

- **Mobile Food Pantry (MFP) Demand Data:** Contains historical demand per location of the MFP in 2019
- **IPUMS USA Data:** Population-level information (*IPUMS USA*, 2025)
- **Local Climatological Data (LCD) Dataset:** Data on weather-patterns (*Local Climatological Data (LCD)*, 2025)
- **Census Bureau:** Includes shapefiles for generating sample households (*Cartographic Boundary Files - Shapefile*, 2018)
- **Census Bureau Geocoding:** API to obtain IPUMS for food banks (*Census Geocoder*, 2025)
- **Harvard School of Public Health, The Nutrition Source:** Healthy Eating Plate

(*Healthy Eating Plate*, 2011)

- **US Department of Agriculture:** Data on Food Group Consumption by Demographics (*Dietary Quality by Food Source and Demographics in the United States*, 2018)

## 2.2 Assumptions

When modeling demand from summary statistics, an assumption was made that the demand was normally distributed. Throughout the report, assumptions are made about the shapes of the distributions motivated by either statistical analysis, or through explained assumptions. For example, when fitting principal components for the Fisher Market Allocation, data was fit to multiple distributions, and the best result was assumed to be the ground truth.

Additionally, when generating sample households, the choice for technical simplicity was made to assume a uniform density, which is not technically accurate considering urban centers and towns. These households also had characteristics generated from independent distributions for each characteristic, which is also not completely true in reality (ex. income and food stamps).

Lastly, for the Fisher Market, the assumption was made that arrivals to the food bank would be distributed according to the types' respective population fractions. While this is obviously not true (very few wealthy people visit food banks), the assumption was made to simplify the problem and to show the extent of the model's capabilities. As shown later on, this assumption ends up not having a significant effect on the final allocation.

## 2.3 Demand Analysis

The first step in the demand analysis was simulating the demand using random normal distribution with the mean and standard deviation from MFP Demand Data (2019). This includes data related to: Density, Household income, Food stamps, Poverty, Number of children, Total population.

Afterwards, the demand data was scaled and Principal Component Analysis was applied. PCA synthesizes the information from large datasets into a smaller set of uncorrelated variables known

as "principal components." This was used to remove correlation between features in MFP Demand Data.

From there, three linear combinations were created from the processed data – explaining  $> 90\%$  of the variance in the demand. When regression was done on the original, the results had a very large condition number (order of  $10^7$ ), hinting at near-multicollinearity. Comparing this to the regression results from PCA, as seen later in the results section, the components were significant.

## 2.4 Sample Household Generation

To obtain a better estimate and sample characteristics of the various regions explored in the dataset, sample households were created using shapefiles from the Census Bureau. Samples were drawn using an acceptance-rejection algorithm.

The samples were drawn uniformly within each PUMA according to its population-weighted density, and characteristics like household size, income, poverty status, access to car, etc. were generated according to parameterized distributions fit on American Community Survey (ACS) data (*IPUMS USA*, 2025).

Households were then typed using the previous PCA analysis, and each food bank was then assigned to (non mutually-exclusively) all households within 25 miles of them to estimate the incidence rates of types and transportation access.

Below is an example showing two food banks within the system and the households generated within the 25 mile distance.

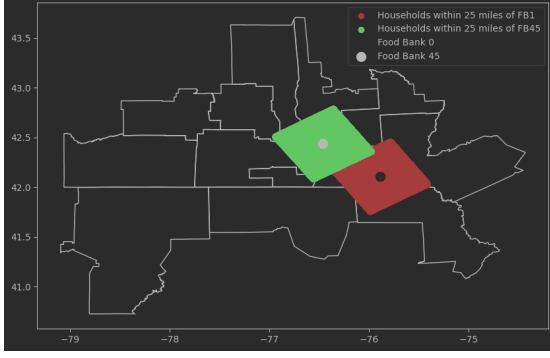


Figure 1: Generated Households within 25 miles of Food Bank 1 and Food Bank 45

## 2.5 Mathematical Modeling

### 2.5.1 Fair Distribution Routing Model

The second task was to develop a model that could create an optimal routing scheme for visiting various food banks throughout the year. This model assumes a flat utility per visit with a non-linear increase per visit, as well as time-invariance for utilities.

The mathematical formulation includes performance metrics for *efficiency* and *fairness*.

#### Sets:

- $i, l \in I$ : set of sites (food banks)
- $j \in J$ : set of days ( $52 \cdot 5 = 260$ )
- $k \in K$ : set of “spots” (visit positions within a day), where  $|K| = 3$

#### Parameters:

- $u_i$ : utility function for visiting site  $i$ , based on demand
- $d_{i,l}$ : distance from site  $i$  to site  $l$  (in miles)
- $\mu$ : maximum distance allowed per day

#### Site Visit Assignment:

$x_{i,j,k} \in \{0,1\}$  for  $i \in I, j \in J, k \in K$ , which equals 1 if site  $i$  is visited on day  $j$  in spot  $k$ ; 0 otherwise.

#### Total Visits per Site:

$$V_i \geq 0 \quad \text{for } i \in I,$$

where

$$V_i = \sum_{j \in J} \sum_{k \in K} x_{i,j,k}.$$

#### Effective (Decayed) Visits:

$$t_i \geq 0 \quad \text{for } i \in I,$$

with the nonlinear constraint:

$$t_i \leq \sqrt{cV_i}$$

This constraint models diminishing returns, implying that the effective benefit from additional visits grows at a sublinear rate.

#### Transition Variables:

$y_{i,l,j,k} \in \{0,1\}$  for  $i, l \in I$  with  $i < l, j \in J, k \in \{0,1\}$ , which equals 1 if, on day  $j$ , site  $i$  is visited in spot  $k$  and site  $l$  is visited immediately after (in spot  $k+1$ ); otherwise, 0.

**Objective Function:** The goal is to maximize the total effective utility from all visits:

$$\text{Maximize: } \sum_{i \in I} u_i t_i.$$

#### Constraints:

##### 1. One Visit per Spot per Day

For every day  $j$  and spot  $k$ :

$$\sum_{i \in I} x_{i,j,k} \leq 1.$$

This ensures that at most one site is visited in a given spot on each day.

##### 2. One Visit per Site per Day

For every site  $i$  and day  $j$ :

$$\sum_{k \in K} x_{i,j,k} \leq 1.$$

This ensures that a site is visited at most once on any day.

##### 3. Linking Transition Variables

For every pair of distinct sites  $i, l$  with  $i < l$ , for every day  $j$ , and for each adjacent spot  $k \in \{0,1\}$ :

$$y_{i,l,j,k} \leq x_{i,j,k},$$

$$y_{i,l,j,k} \leq x_{l,j,k+1},$$

$$y_{i,l,j,k} \geq x_{i,j,k} + x_{l,j,k+1} - 1.$$

These constraints guarantee that  $y_{i,l,j,k} = 1$  if and only if site  $i$  is visited in spot  $k$  and site  $l$  is visited in the subsequent spot  $k+1$  on day  $j$ .

##### 4. Daily Travel Distance Limit

For every day  $j$ :

$$\sum_{i \in I} \sum_{\substack{l \in I \\ l > i}} \sum_{k \in \{0,1\}} d_{i,l} y_{i,l,j,k} \leq \max\_miles.$$

This ensures that the sum of the distances for consecutive visits on any given day does not exceed the maximum allowed distance.

## 5. Total Visits Definition

For every site  $i$ :

$$V_i = \sum_{j \in J} \sum_{k \in K} x_{i,j,k}.$$

### Optimization Problem:

$$\text{maximize } \sum_{i \in I} u_i t_i$$

$$\text{subject to } t_i \leq \sqrt{c V_i}, \quad \forall i \in I,$$

$$V_i = \sum_{j \in J} \sum_{k \in K} x_{i,j,k}, \quad \forall i \in I,$$

$$\sum_{i \in I} x_{i,j,k} \leq 1, \quad \forall j \in J, \forall k \in K,$$

$$\sum_{k \in K} x_{i,j,k} \leq 1, \quad \forall j \in J, \forall k \in K,$$

$$\sum_{i \in I} \sum_{l < i} \sum_{k \in K} d_{i,l} y_{i,l,j,k} \leq \mu, \quad \forall j \in J,$$

$$y_{i,l,j,k} \leq x_{i,j,k}, \quad \forall i, l \in I, \forall j \in J, \forall k \in K,$$

$$y_{i,l,j,k} \leq x_{l,j,k}, \quad \forall i, l \in I, \forall j \in J, \forall k \in K,$$

$$y_{i,l,j,k} \leq x_{i,j,k} + x_{l,j,k} - 1, \quad \forall i, l \in I, \forall j \in J, \forall k \in K,$$

$$x_{i,j,k} \geq 0, \quad \forall i \in I, \forall j \in J, \forall k \in K.$$

### Utilitarianism (Efficiency)

$$\text{Total utility} = \sum_{i \in I} V_i u_i$$

vs.

$$\text{Total utility (2019)} = \sum_{i \in I} V_i^{2019} u_i$$

### Equality Gap (Fairness)

$$\text{Equality Gap} = \sum_{i \in I} \min(0, V_i u_i - V_i^{2019} u_i)$$

**Coefficient of Variation (Fairness)** Let  $\mu, \sigma$  be the sample mean and sample standard

deviation of utilities in an allocation. Then the coefficient of variation is  $\frac{\mu}{\sigma}$ .

**Gini Coefficient (Fairness)** Let  $\mu$  be the sample mean of the utilities. Then:

$$\frac{\sum_{i=1}^n \sum_{j=1}^n |u_i - u_j|}{2n^2 \mu}$$

### Interpretation

- **Decision Variables  $x_{i,j,k}$ :** Assign which site is visited on which day and in which time slot (spot).
- **Transition Variables  $y_{i,l,j,k}$ :** The transition variables track the sequence of visits within the same day, making sure to compute travel distance between consecutive sites.
- **Effective Visits  $t_i$ :** The quadratic constraint  $t_i^2 \leq c V_i$  captures the idea of diminishing marginal return as a site is visited more times.
- **Travel Constraint:** The sum of the distances between consecutively visited sites on any day is limited to the max distance.
- **Evaluation Metrics (Utilitarianism):** One metric was constructed to evaluate the efficiency characteristic of the solution. The rationale for this is that utilitarianism is Pareto Efficient where utility is quasi-linear with monetary transfers. Since the food allocation can be expressed in dollar terms it's a reasonable assumption here.
- **Evaluation Metrics (Fairness):** Three metrics were constructed to evaluate the fairness characteristic of the solution. The efficiency gap is the total amount of utility needed to be reallocated to the sites that lose utility in order for the current solution to make no site worse off. This metric is closely related to Pareto Efficiency as it's the utility value of the monetary transfers needed to construct a Pareto improvement. The coefficient of variation measures of the relative spread of the utilities, and the gini coefficient is widely used to measure economic inequality.

## 2.5.2 Cold-weather Routing Model

The mathematical modeling remains the same for cold weather. However, there are slight changes

to account for a reduced number of served sites. Instead of the seventy locations, the new mathematical model will instead restrict the total number of sites that can be visited, hopefully better optimizing to meet the adjusted demand. Additionally, demand is scaled by the fraction of households within range of the food bank that have a car, representing the fact that without a car it would be difficult to visit in winter. Other types of transportation (bus, bicycle, train etc.) were considered, but were so low as to be negligible.

Since access to these locations changes, how this is incorporated into efficiency and what will happen to the equality gap will be discussed. The following changes are made to the previous model:

#### Sets:

- $i \in I$ : set of sites (food banks).
- $j \in W$ : set of winter days (64 total).
- $k \in K$ : set of “spots” (visit positions within a day), where  $|K| = 3$ .

Here the set of days is simply the winter days, and so this model solves a *sub-problem* over only the 64 winter days. **Decision Variable:**

$x_{i,j,k}$  indicates whether site  $i$  is assigned on day  $j$  in slot  $k$  (e.g. binary or nonnegative).

#### Added Winter Constraint:

$$x_{i,j,k} \leq Mz_i \quad \forall i \in I, j \in J, k \in K,$$

$$\sum_{i \in I} z_i \leq 20,$$

where  $W \subseteq J$  is the set of winter days,  $x_{i,j,k}$  indicates whether site  $i$  is visited on day  $j$  in slot  $k$ , and  $M$  is a sufficiently large constant to ensure  $x_{i,j,k} = 0$  unless  $z_i = 1$ . This forces the model to select the best subset of sites for winter distribution. The same efficiency and fairness metrics introduced earlier are used to evaluate performance under these winter constraints.

#### Efficiencies (Total Utility)

$$\text{Total Utility} = \sum_{i \in I} V_i u_i$$

vs.

$$\text{Total Utility (2019)} = \sum_{i \in I} V_i^{2019} u_i$$

**Fairness Metrics** The fairness metrics here are the same as in the previous problem, only benchmarked against two allocations: the unconstrained allocation and the 2019 allocation. The unconstrained allocation is the previous model without the site constraint, essentially the original standard model solved over the winter sub-problem. The 2019 allocation is the fraction of the year that winter is, multiplied by the number of visits in 2019, giving the ‘winter visits’ in 2019.

#### 2.5.3 Fisher Market Allocation

For a stochastic demand with a fixed capacity  $C_k$  for resources, and a fixed set of locations, the model requires a new algorithm to address this challenge. To do fair resource allocation, an Eisenberg-Gale formulation with a Cobb-Douglas utility function for dairy, greens, protein, and grains will be used. The function will take as input the amount of resources remaining (i.e. the budget), the total number of future arrivals of each type, as well as the weights for the linear utility assumption. It will then return the optimal allocation as well as its objective value.

The final model is one that allocates a set of four resources ( $k$ ) over a set of fixed sites ( $t$ ). The fixed route includes 12 sites with the highest average demand, and the four resources were derived from the USDA eating guidance and include: dairy, fruits and vegetables, grains, and protein. The fixed capacity was chosen to be 180,000 lbs to make the problem feasible.

Additionally, a set of household types  $\theta$  were constructed based on the results of the PCA analysis earlier.

#### 1. PCA Transformation, Distribution Fitting

The first step was converting the correlation heatmap (covering household income, density, food stamps, poverty, number of children, and total population) into a numeric matrix, applying Principal Component Analysis (PCA).

- Different distributions (normal, lognormal, exponential, Pareto) were tested for each principal component (PC) to identify the best fit.

- From these tests, the conclusions are:
  - PC1 follows a Pareto distribution
  - PC2 follows a Beta distribution
  - PC3 follows a Normal distribution

2. **Household Type Classification** Once the principal components were identified, the components are split into three quantiles (0–33%, 33–66%, 66–100%), labeling them as 0 (low), 1 (medium), or 2 (high). Thus, each household is classified by a “type code” (PC1,PC2,PC3). For example:

- A household with type (0,2,1) is in the low group for PC1, high group for PC2, and middle group for PC3.
- However, as seen later, all households fall into category 2 for PC2, and so there are 9 types instead of 27.

### 3. Utility Weights ( $\alpha$ Parameters).

With the American Community Survey (ACS) data for consumption patterns by income and age,  $\alpha$ -weights are assigned in a Cobb-Douglas utility function for each resource (dairy, fruits and vegetables, grains, and protein). To do so, data from the US Department of Agriculture (*Dietary Quality by Food Source and Demographics in the United States*, 2018) is used to generate synthetic data based on income and age levels from summary statistics, and the synthetic data is projected onto the same principal component space. This allows for the classification of the synthetic data into household types, enabling four separate regressions to be run, where the response is a resource and the features are types. To capture the effect that types with lower consumption of a resource would extract more utility, a model for  $\alpha_{type,resource}$  is taken to be  $\alpha = \frac{1}{\hat{\beta}_{type,resource}}$ , such that  $\hat{\beta}$  is the regression coefficient. These values are then scaled to have a sum of 1 to comply with the Cobb-Douglas formulation, which captures a percentage-wise increase in all inputs producing the same effect size on the output. Further, a parameter  $A_\theta$  is defined by an examination of the types. For example, when investigating the principal components, it can be seen that the first principal com-

ponent is household income, therefore a low  $A_\theta \approx 0$  is a sensible choice to model low utility for very high income families ( $\theta = 2xx$ ).

#### Sets:

- $t \in T$ : set of sites.
- $\theta \in \Theta$ : set of family types (or site types).
- $k \in K$ : set of resource types (e.g. dairy, produce, grains, etc.).

#### Parameters:

- $c_{t,k}$ : the capacity (supply) of resource  $k \in K$  during period  $t \in T$ .
- $N_{t,\theta}$ : the number of families (or demand units) of type  $\theta$  expected in period  $t$ .
- $\alpha_{\theta,k}$ : Cobb–Douglas weight of family type  $\theta$  for resource  $k$ .
- $A_\theta$ : scale factor for type  $\theta$  (e.g. reflecting overall importance or base consumption).

#### Decision Variables:

$$x_{t,\theta,k} \geq 0, \quad \forall t \in T, \theta \in \Theta, k \in K,$$

representing the amount of resource  $k$  allocated to each family of type  $\theta$  at site  $t$ .

#### Utility Function (Cobb–Douglas):

For each type  $\theta$ , define

$$u_\theta(x_{t,\theta}) = \prod_{k \in K} (x_{t,\theta,k})^{\alpha_{\theta,k}}.$$

Taking logarithms turns this product into a sum of  $\alpha_{\theta,k} \log(x_{t,\theta,k})$ .

#### Objective Function:

$$\max_{\{x_{t,\theta,k}\}} \sum_{\theta \in \Theta} A_\theta \sum_{t \in T} N_{t,\theta} \log(u_\theta(x_{t,\theta,k}))$$

#### Constraints:

$$\sum_{\theta \in \Theta} [N_{t,\theta} x_{t,\theta,k}] \leq c_k \quad \forall t \in T, k \in K,$$

$$x_{t,\theta,k} \geq 0 \quad \forall t, \theta, k.$$

#### Interpretation:

- The objective function maximizes the utility across all the different periods  $t$  as well as family type  $\theta$ .



- The capacity constraint makes sure that for each resource  $k$ , the total allocated does not exceed the available capacity  $c_{t,k}$  in period  $t$ .
- The nonnegative constraint makes sure there isn't a nonnegative allocations.
- The number of families is not deterministic in this model and so a certainty equivalence procedure was implemented. The procedure works as follows:
  - Move to site  $t$  and realize arrivals  $N_{t,\theta}$
  - All future sites have arrivals set to their expectation
  - All future sites have arrivals set to their expectation  $[N_{t',\theta}]$  for  $t' > t$
  - An allocation  $x_{t,\theta,k}$  is generated
  - The total resources allocated at that step  $\sum_{\theta} N_{t,\theta} x_{t,\theta,k}$  is subtracted from the capacity  $c_k$
  - The process is repeated at the next site with the updated capacity until all sites have been allocated to

### Hindsight Gap (Efficiency)

Let  $u_{myopic}$ ,  $u_{hindsight}$  be the total utility of the myopic and hindsight solutions, respectively. Then the hindsight gap is:

$$\frac{u_{hindsight} - u_{myopic}}{u_{hindsight}}$$

The hindsight gap measures how much better the hindsight solution is compared to the myopic solution.

### Maximum Envy (Fairness)

Let  $u_{t,\theta} = \sum_k u_{t,\theta,k}$  be the total utility at each site  $t$  for each type  $\theta$ . Then the maximum envy for each type is:

$$E_{\theta} = \max(u_{t',\theta} - u_{t,\theta}, \forall t, t' \neq t)$$

The total maximum envy is:

$$\sum_{\theta} E_{\theta}$$

The maximum envy measures how much a given household type could improve by being at a different site, and the aggregate metric is the total amount of 'envy' across the whole model.

## 3 Results

### 3.1 Demand Analysis

Having outlined the approach to simulating the demand using a random normal distribution, now the key findings are presented below.

First, as shown below, the variance captured in different numbers of PCA components is explored. Next, Principal Component Analysis (PCA) is explained as reducing the dimensionality of these features, revealing the core factors driving demand. Finally, its discussed how these insights shape subsequent allocation and routing decisions in the model.

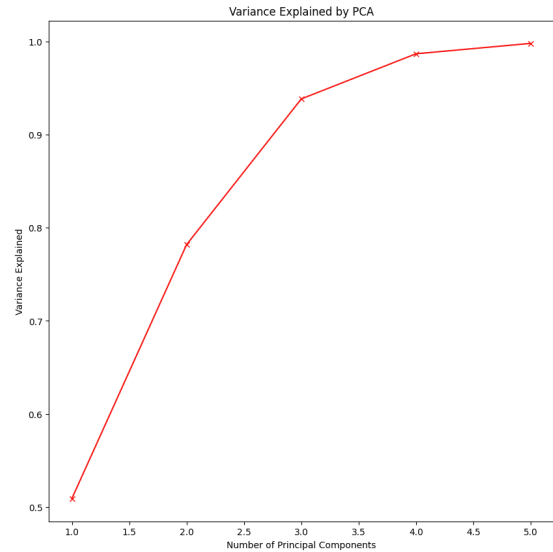
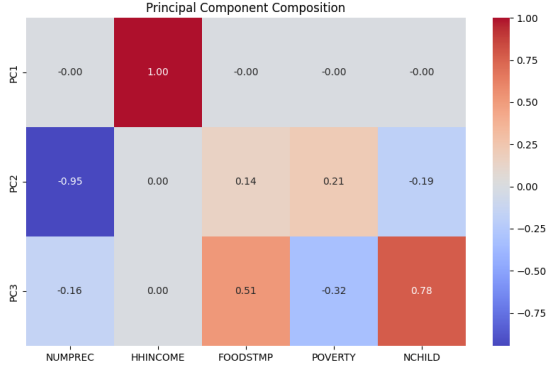


Figure 2: Variance Explained by PCA (no y column)

As seen, PCA is able to explain more than 90% of the variance in the demand in 3 dimensions. The heatmap below shows how these original features contribute to each principal component.



Regression on the demand per year helps show which factors are most important in predicting this metric:

Figure 3: Heatmap describing the three principal component features

| Variable | Coef.     | Std. Err. | t      | p-value | [0.025   | 0.975]  |
|----------|-----------|-----------|--------|---------|----------|---------|
| const    | -171.7447 | 63.886    | -2.688 | 0.007   | -297.172 | -46.318 |
| NUMPREC  | 127.8586  | 64.644    | 1.978  | 0.048   | 0.944    | 254.773 |
| DENSITY  | -0.0507   | 0.014     | -3.542 | 0.000   | -0.079   | -0.023  |
| HHINCOME | -0.0023   | 0.002     | -1.188 | 0.235   | -0.006   | 0.001   |
| FOODSTMP | -5.4136   | 11.994    | -0.451 | 0.652   | -28.961  | 18.134  |
| POVERTY  | -1.6749   | 0.959     | -1.747 | 0.081   | -3.557   | 0.207   |
| NCHILD   | 248.3854  | 71.665    | 3.466  | 0.001   | 107.688  | 389.083 |
| PUMA_pop | 0.0041    | 0.001     | 3.401  | 0.001   | 0.002    | 0.007   |

Table 1: OLS Regression Results (on Normal Space)

Looking at the regression results from this table, there are clearly non-significant features like food-stamps, household income and poverty. This issue, in light of the condition number of

the regression in the order of  $10^7$  hints at near-multicollinearity, necessitating the use of PCA to reduce feature covariance.

| Variable | Coef.     | Std. Err. | z      | p-value | [0.025   | 0.975]   |
|----------|-----------|-----------|--------|---------|----------|----------|
| const    | 143.2850  | 3.479     | 41.192 | 0.000   | 136.467  | 150.103  |
| PC1      | -0.0017   | 0.000     | -4.057 | 0.000   | -0.002   | -0.001   |
| PC2      | -180.8965 | 31.031    | -5.830 | 0.000   | -241.716 | -120.077 |
| PC3      | -626.5524 | 176.689   | -3.546 | 0.000   | -972.856 | -280.249 |

Table 2: OLS Regression Results (on PCA Space)

Comparing this to the regression results from PCA, all components are significant. Further, a shrinkage to an order of  $10^5$  can be observed in the condition number, which improves the multicollinearity problem.

Combining the regression with the heatmap, the reader can see how socioeconomic factors contribute to the demand analysis. For example, the first principal component is purely household income, where the regression coefficients have a neg-

ative effect on the demand. Whereas, the second principal component is almost purely the negative number of people in the household, which when multiplied by the regression coefficient, shows a very significant contribution to demand as the number of people increase. The third principal component is the miscellaneous/composite feature that is firstly concerned with the number of children, but contain cross terms. While this term does not have a direct description, it is very significant nevertheless, unifying the demand regression.

## 3.2 Proposed Schedule

### 3.2.1 Standard Routing Model

| Day                         | Site 1                                | Site 2                                | Site 3                           |
|-----------------------------|---------------------------------------|---------------------------------------|----------------------------------|
| 1                           | MFP Waverly                           | MFP Senior – Addison Place Apartments | MFP Lansing                      |
| 2                           | MFP Birnie Transportation Services    | MFP Tuscarora                         | MFP Owego VFW                    |
| 3                           | MFP Van Etten                         | MFP Richford                          | MFP College TC3 -College         |
| 4                           | MFP College Corning Community College | MFP The Love Church                   | MFP First Assembly Of God Church |
| 5                           | MFP Woodhull                          | MFP Waverly                           | MFP Campbell                     |
| 6                           | MFP Windsor                           | MFP Waverly                           | MFP Salvation Army Ithaca        |
| 7                           | MFP College Corning Community College | MFP Senior - Dayspring                | MFP Redeemer Lutheran Church     |
| 8                           | MFP College Corning Community College | MFP The Love Church                   | MFP Lansing                      |
| 9                           | MFP Wayland                           | MFP Rathbone                          | MFP Redeemer Lutheran Church     |
| 10                          | MFP Redeemer Lutheran Church          | MFP Owego VFW                         | MFP Schuyler Outreach            |
| 11                          | MFP Whitney Point                     | MFP Lindley                           | MFP Erin                         |
| 12                          | MFP Lamphear Court                    | MFP Boys and Girls Club               | MFP Lindley                      |
| 13                          | MFP Woodhull                          | MFP Van Etten                         | MFP American Legion - Binghamton |
| 14                          | MFP Rathbone                          | MFP Richford                          | MFP Salvation Army Ithaca        |
| ⋮ (continues until day 260) |                                       |                                       |                                  |

Table 3: Proposed MFP Schedule (First 14 Days out of 260)

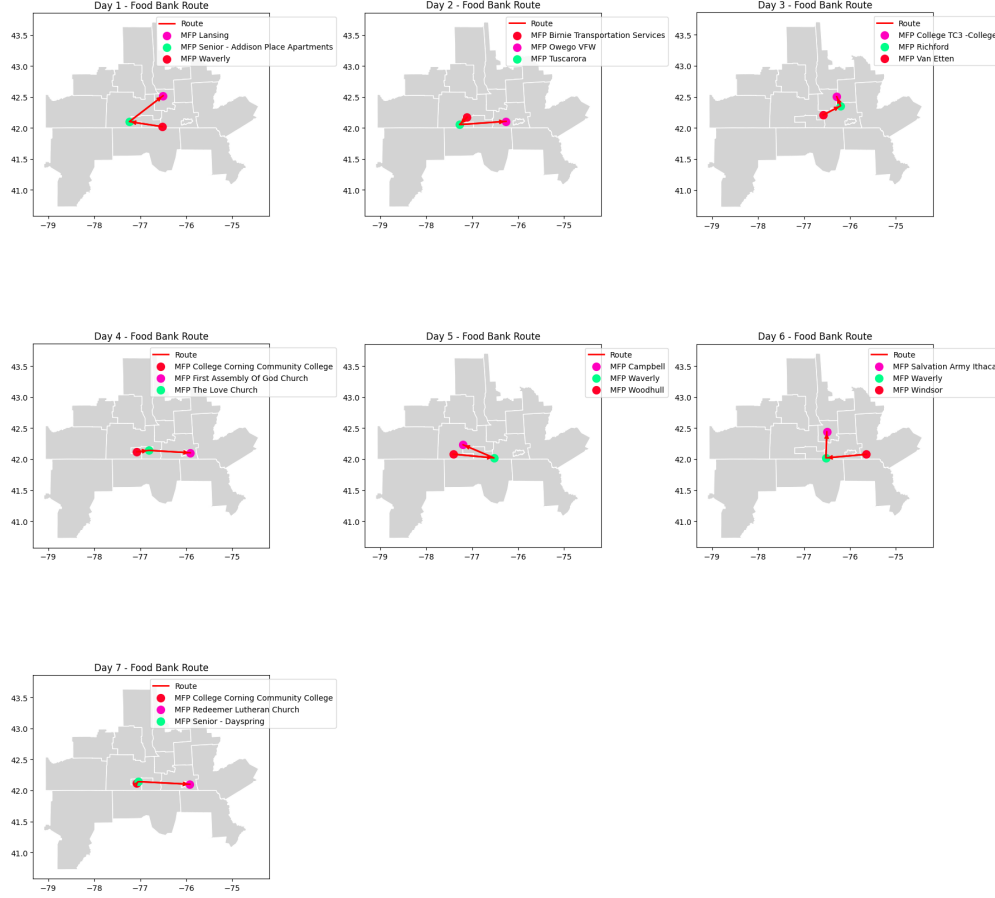


Figure 4: Food Bank Route Visualization (First 7 of 260 Days)

### 3.2.2 Winter Routing Model

| Day                         | Site 1                                | Site 2                                | Site 3                              |
|-----------------------------|---------------------------------------|---------------------------------------|-------------------------------------|
| 1                           | MFP Van Etten                         | MFP Deposit                           | MFP American Legion - Binghamton    |
| 2                           | MFP Van Etten                         | MFP Tuscarora                         | MFP Avoca                           |
| 3                           | MFP Lansing                           | MFP Beaver Dams                       | MFP American Legion - Binghamton    |
| 4                           | MFP Van Etten                         | MFP Senior - Woodsedge Apartments     | MFP Avoca                           |
| 5                           | MFP Lansing                           | MFP Deposit                           | MFP Birnie Transportation Services  |
| 6                           | MFP Senior - Woodsedge Apartments     | MFP Deposit                           | MFP College TC3 -College            |
| 7                           | MFP Whitney Point                     | MFP Prattsburgh                       | MFP Endwell United Methodist Church |
| 8                           | MFP Van Etten                         | MFP Colesville                        | MFP Boys and Girls Club             |
| 9                           | MFP Senior - Woodsedge Apartments     | MFP Reach for Christ Church Freeville | MFP American Legion - Binghamton    |
| 10                          | MFP Van Etten                         | MFP Redeemer Lutheran Church          | MFP Lindley                         |
| 11                          | MFP Van Etten                         | MFP Redeemer Lutheran Church          | MFP College TC3 -College            |
| 12                          | MFP Reach for Christ Church Freeville | MFP Rehoboth Deliverance Ministry     | MFP Bath                            |
| 13                          | MFP Van Etten                         | MFP Canistota                         | MFP Avoca                           |
| 14                          | MFP Whitney Point                     | MFP Rehoboth Deliverance Ministry     | MFP Birnie Transportation Services  |
| ⋮ (continues until day 260) |                                       |                                       |                                     |

Table 4: Proposed MFP Schedule (First 14 Days out of 260)

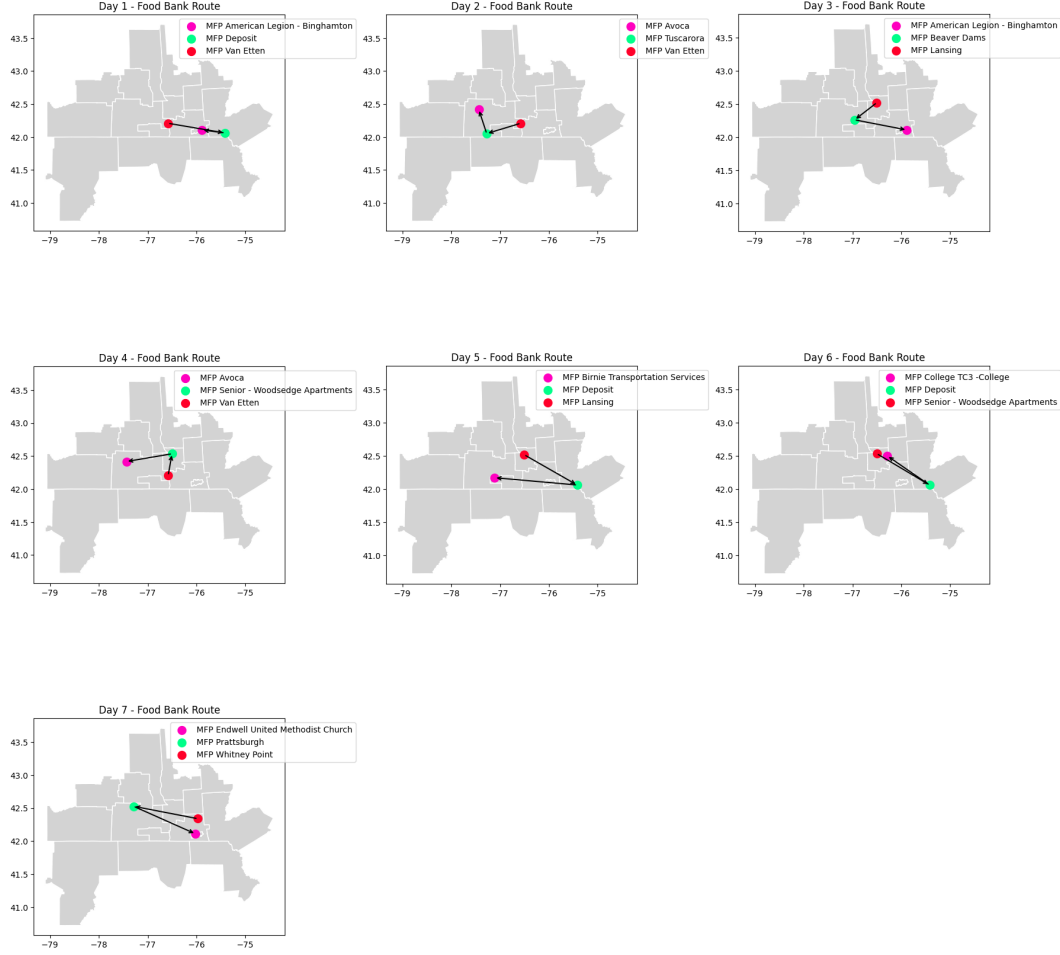


Figure 5: Food Bank Route Visualization (First 7 Days out of 64 in Winter)

### 3.3 Type Formulations and Regressions

#### 3.3.1 Distributions of Principal Components

In examining the best fitting distribution for the principal component data for the ACS dataset, the candidate distributions were Normal, Lognormal, Beta, Pareto and Exponential. By comparing QQ plots, it is possible to gauge the best fitting distribution, adopting simpler distributions when possible. Below are the best possible distributions for each principal component.

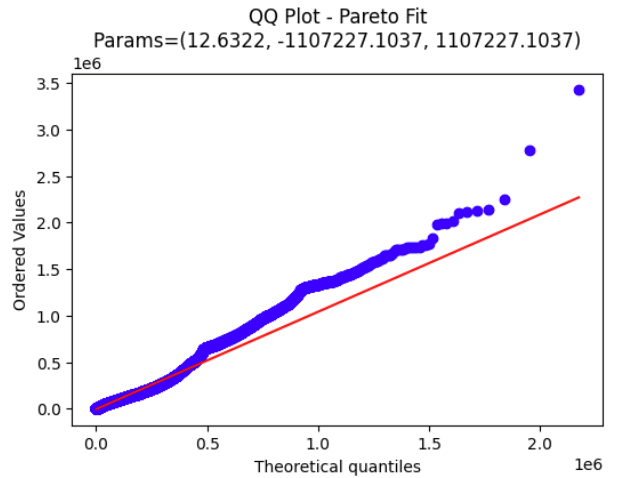


Figure 6: Pareto Fit for the First Principal Component

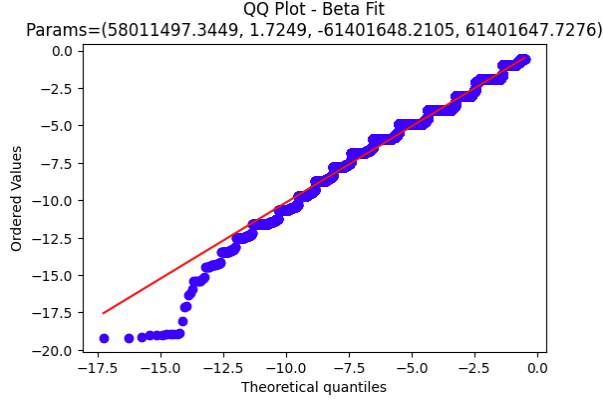


Figure 7: Beta Fit for the Second Principal Component

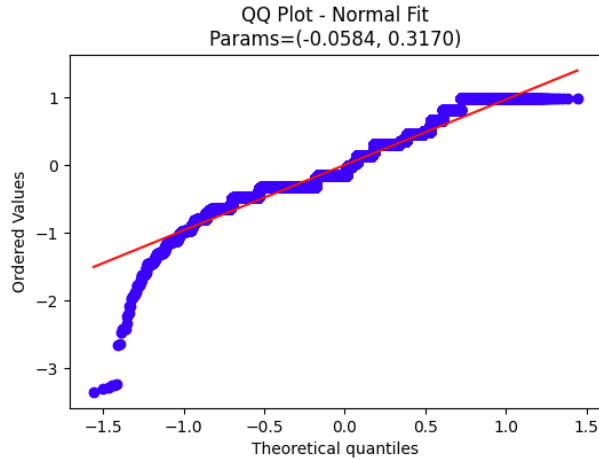


Figure 8: Normal Fit for the Third Principal Component

As an example, the Pareto distribution could easily be rejected for the second principal component, by virtue of the clear divergence shown below.

While these fits are not perfect, they offer better explanations of the data compared to all other distributions. The best parameters were chosen by a solver, and QQ plots alongside visual judgment was preferred over Kolmogorov-Smirnov Goodness of Fit tests due to abundance of data ( $> 600000$ ).

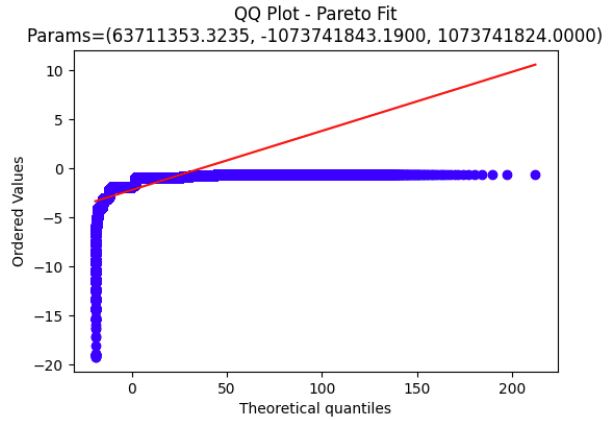


Figure 9: Pareto Fit for the Second Principal Component

### 3.3.2 Regressions on Resources by Type

As expected, high income types are likelier to have greater consumption as a direct consequence of their incomes. The regression results for dairy products are as presented in Table 5.

| Variable | Coef.  | Std. Err. | t       | p-value | [0.025 | 0.975] |
|----------|--------|-----------|---------|---------|--------|--------|
| 020      | 3.2100 | 0.223     | 14.373  | 0.000   | 2.772  | 3.648  |
| 021      | 3.7865 | 0.169     | 22.369  | 0.000   | 3.455  | 4.118  |
| 022      | 3.4470 | 0.076     | 45.405  | 0.000   | 3.298  | 3.596  |
| 120      | 3.3181 | 0.132     | 25.090  | 0.000   | 3.059  | 3.577  |
| 121      | 4.0861 | 0.108     | 37.815  | 0.000   | 3.874  | 4.298  |
| 122      | 3.7132 | 0.051     | 72.542  | 0.000   | 3.613  | 3.813  |
| 220      | 4.1594 | 0.089     | 46.595  | 0.000   | 3.984  | 4.334  |
| 221      | 4.7933 | 0.066     | 72.764  | 0.000   | 4.664  | 4.922  |
| 222      | 4.1632 | 0.035     | 118.262 | 0.000   | 4.094  | 4.232  |

Table 5: OLS Regression Results For Dairy Products by Type

This regression was run three additional times methodology section, the final utility weights are for each food group, where every feature was  $p=0$ . as in Table 6. By applying the transformations described in the

| type | A    | $\alpha_{dairy}$ | $\alpha_{greens}$ | $\alpha_{protein}$ | $\alpha_{grains}$ |
|------|------|------------------|-------------------|--------------------|-------------------|
| 220  | 0.05 | 0.444543         | 0.317092          | 0.129595           | 0.108771          |
| 021  | 1.00 | 0.437737         | 0.326800          | 0.132201           | 0.103263          |
| 022  | 1.00 | 0.444040         | 0.323506          | 0.129201           | 0.103253          |
| 020  | 1.00 | 0.432452         | 0.329777          | 0.134860           | 0.102911          |
| 122  | 0.60 | 0.446640         | 0.319025          | 0.128491           | 0.105844          |
| 221  | 0.05 | 0.453017         | 0.312245          | 0.125997           | 0.108740          |
| 120  | 0.40 | 0.430723         | 0.327354          | 0.136242           | 0.105681          |
| 121  | 0.50 | 0.442827         | 0.320828          | 0.130385           | 0.105961          |
| 222  | 0.10 | 0.455311         | 0.310953          | 0.124932           | 0.108803          |

Table 6: Cobb Douglas Weights

### 3.4 Algorithm Evaluation

Comparing the model schedule with benchmark approach using efficiency and fairness definitions.

#### 3.4.1 Fair Distribution Routing Model

**Efficiencies (Total Utility):**

$$\sum_{i \in I} V_i u_i = 176,754.2$$

$$\text{Efficiency (2019)} = \sum_{i \in I} V_i^{2019} u_i = 102,474.4$$

**Equality Gap:**

$$\sum_{i \in I} \min(0, V_i u_i - V_i^{2019} u_i) = -13,712.2$$

**Alternative Equality Metrics:**

$$\text{Coefficient of variation} = \frac{\sigma^*}{\mu^*} = 1.52$$

$$\text{Coefficient of variation (2019)} = \frac{\sigma^{2019}}{\mu^{2019}} = 0.67$$

$$\text{Gini coefficient} = 0.65$$

$$\text{Gini coefficient (2019)} = 0.37$$

Looking at the results of the efficiency and fairness indicators, it becomes clear that the model is more efficient, and less fair than the route from 2019.

Beginning with the efficiency. The efficiency metric is to sum the utility across all days and visits. As shown above, the model captures roughly 70,000 more demand per year than the route from 2019. However, comparing this to the equality gap, it can be seen that the model becomes less fair, as the equality gap for fairness has a gross loss of 13,712.20 suffered by sites that aren't receiving an allocation.

Essentially, the model is visiting sites with a higher demand, or higher utility, more often and ignoring sites with lower demand. The coefficient of variation and gini coefficient are also higher for the current routing model than in 2019, which agree that while it is more efficient, it suffers by being less fair.

### 3.4.2 Winter Routing Model

#### Efficiencies (Total Utility):

$$\begin{aligned} \text{Efficiency (constrained allocation)} \\ &= \sum_{i \in I} V_i^{20} u_i = 51,259.8 \\ \text{Efficiency (unconstrained allocation)} &= \sum_{i \in I} V_i u_i = 43,959.2 \\ \text{Efficiency (2019)} &= \sum_{i \in I} V_i^{2019} u_i = 25,224.47 \end{aligned}$$

#### Equality Gap:

$$\begin{aligned} \text{Equality Gap (constrained vs unconstrained)} \\ &= \sum_{i \in I} \min(0, V_i^{20} u_i - V_i u_i) = -12,704.86 \\ \text{Equality Gap (unconstrained vs 2019 allocation)} \\ &= \sum_{i \in I} \min(0, V_i u_i - V_i^{2019} u_i) = -3,326.38 \end{aligned}$$

#### Alternative Equality Metrics:

$$\begin{aligned} \text{Coef. variation (constrained)} &= \frac{\sigma^{20}}{\mu^{20}} = 2.10 \\ \text{Coef. variation (unconstrained)} &= \frac{\sigma^*}{\mu^*} = 1.50 \\ \text{Coef. variation (2019)} &= \frac{\sigma^{2019}}{\mu^{2019}} = 0.67 \end{aligned}$$

$$\begin{aligned} \text{Gini coefficient (constrained)} &= 0.80 \\ \text{Gini coefficient (unconstrained)} &= 0.64 \\ \text{Gini coefficient (2019)} &= 0.37 \end{aligned}$$

Over winter, the constrained allocation has the highest utility as expected since the model will focus on serving high demand sites first. By constraining it to only choose 20 locations, the utility will be high. The unconstrained allocation does worse with total utility/efficiency, however, the fairness of this allocation is better than the constrained and closer to that of the 2019 allocation.

The constrained allocation performs even worse in winter on the fairness metrics, a Gini coefficient of 0.8 is above even the most unequal societies on Earth. The efficiency gap is almost as large as the full-year allocation previously. This goes down on the unconstrained model to -3,326.38, with a Gini coefficient of 0.64, similar to the fair distribution routing model.

This model is balances between the highest optimal utility and visiting different sites to balance fairness. The next model, the Fisher Market stochastic allocation, attempts to find an equilibrium between these two.

### 3.4.3 Fisher Market / Eisenberg-Gale Algorithm

#### Efficiencies (Total Utility):

$$\begin{aligned} u_{hindsight} &= 5,100.74 \\ u_{myopic} &= 341.90 \end{aligned}$$

#### Hindsight Gap (Efficiency)

$$u_{hindsight} - u_{myopic} = 4,758.8$$

$$\frac{u_{hindsight} - u_{myopic}}{u_{hindsight}} = 93.3\%$$

#### Maximum Envy (Fairness)

The myopic solution performed moderately well with a total maximum envy of 85.60297. However, there is significant variation across types.



Types 2xx (types with the highest income) have 0 envy since they receive the lowest allocation everywhere, while Types 0xx (types with the lowest income) have the greatest envy due to variation in allocation. The hindsight solution has no envy since it allocates equally across all sites.

Comparing the hindsight solution to the myopic solution, the myopic performs much worse than the hindsight with a gap ratio of 93.3%. This large difference can be attributed to the large standard deviations in the arrival, as coefficients of variation for arrival types range from 1.95 to 5.55, which indicate an extremely variable distribution that is difficult to model.

Since the demand model relies on 3D data and is stochastic, the solutions can only be generated with the full dataset. These detailed results can be found in the code files.

#### 4 Discussion & Conclusion

Overall, this analysis indicates that leveraging mathematical modeling and data-driven scheduling can significantly enhance food distribution plans in the face of data variability and fairness constraints. The joint approach—combining demand analysis optimized routing plus efficiency-fairness evaluation—has been able to reveal the complex inter-linkages between socio-economic factors and service delivery, and has generated a model that is far superior to the traditional methods, such as the 2019 baseline.

Despite this, while the proposed schedule has achieved greater overall efficiency through the extraction of 74,279.8 additional utility from the food distribution compared to that of 2019, the

equality gap loss of 13,712.2 indicates the continued challenge of achieving equitable access. The 0.85 increase in the coefficient of variation as well as the 0.28 increase in the Gini coefficient further emphasize this struggle with equality. Based on these a result, a clear efficiency-fairness tradeoff is present.

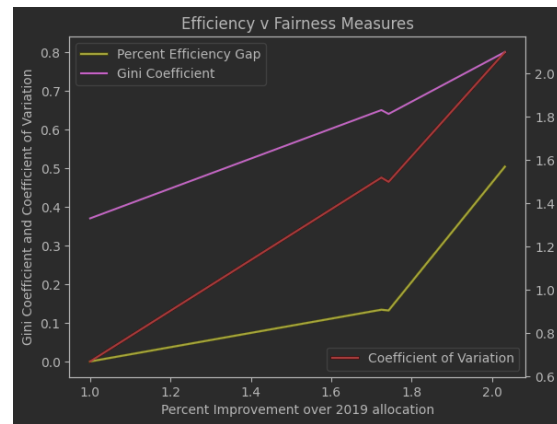


Figure 10: Efficiency versus Fairness Metrics

This is a crucial point for determining the purpose of the food bank itself. Operators, donors and policymakers must ask if the goal is to serve the most people possible or to ensure as broad coverage as possible. Different stakeholders will have different views on these issues, and like most policy problems, there is no clear-cut correct answer. Any advancement on one front will almost surely require giving ground on another.

Ultimately, this study not only advances the understanding of the logistics of food distribution but also offers a solid foundation on which to make adaptive, data-driven plans to enhance food security among under-served populations.

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